which contained the fields *userID*, *movieID* and *value*. Here, *userID* and *movie ID (itemID)* refer to a particular user and a particular item, and *value* denotes the strength of the interaction (e.g., the rating given to a movie).

The following steps can be used to run the Recommendation algorithm.

Create a directory in the Hadoop file system to store the ratings file using the following command:

```
hadoop fs -mkdir /mahout_data/
```

Copy the downloaded file to HDFS using the between the following command:

```
hadoop fs -put /home/hduser/mydata/ml-latest/
ratings.csv /mahout_data/
```

Go to the Mahout directory, cd */usr/local/mahout/bin/* and issue the following command (the output file should be unique and JAVA_HOME should be properly set):

```
./mahout recommenditembased -s
SIMILARITY_LOGLIKELIHOOD -i hdfs://
localhost:9000/mahout_data/ratings.csv
-o hdfs://localhost:9000/ratings_test/
--numRecommendations 25
-i hdfs://localhost:9000/mahout_data/ratings.csv - Denotes
the input file
-o hdfs://localhost:9000/ratings_test/  -denotes the
output file .
```

The command will run for a couple of minutes and you can see your output from the Web interface as well, as shown in Figure 2.

Note: In the above snippet, *recommenditembased* means we are creating an item based recommendation and not a user based recommendation. The difference between the two is that a user based recommendation finds similar users based on what they like, and item based recommendation figures out what the user likes and finds items to match those preferences. Mahout's item-based recommendation algorithm takes customer preferences by item as input, and generates an output recommending similar items with a score indicating whether a customer will *like* the recommended item.

You can check the output file and it will contain two columns: the *userID* and an array of *itemIDs* and scores. This could, for instance, recommend a user's preference for a particular movie, which he may be interested in watching.

Figure 1: Building Apache Mahout on Ubuntu

Figure 2: Browsing directory

"We are leaving the age of information and entering the age of recommendation," says Chris Anderson, British entrepreneur and the curator of TED. The increasing adoption of the Web as a vehicle for business has changed the way in which businesses interact with the customer. Marketing teams and businesses are now using intelligent algorithms and social technologies to form meaningful, ongoing relationships with customers. It is here that technologies like Mahout Recommender will play a key role.

References

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